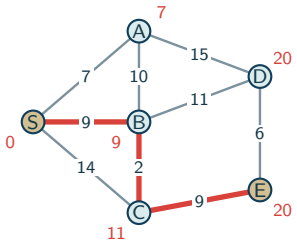
GRAPH THEORY LADDER 9 OF 9

Weighted networks: shortest paths and spanning trees

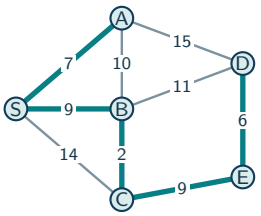
Costs on the edges give practical questions, some with fast answers and some with none.

1 / THE RULE AND THE PICTURE

The shortest route and the cheapest network are different trees



shortest route $S \to E$: 20



minimum spanning tree: 33

Each edge carries a weight such as a length or a cost, and two questions arise: the shortest route from S, solved by Dijkstra, and the cheapest set of edges joining every vertex, solved by Kruskal. The distances from S are in red.

2 / DIJKSTRA'S METHOD

Settle the nearest unsettled vertex and update its neighbours

settled	distance	reached
S	0	start
A	7	from S
B	9	from S
C	11	from B $(9+2)$
D	20	from B $(9+11)$
E	20	from C $(11+9)$

Starting from S the vertices are settled in order of distance, and each distance can only be improved through vertices already settled: C is reached at $9+2=11$ through B, beating the direct edge of 14, and E costs $11+9=20$ through C, beating $20+6=26$ through D.

3 / KRUSKAL'S METHOD

Take the cheapest edge that does not close a cycle

Sorting the edges by weight, $BC=2$, $DE=6$, $SA=7$, $SB=9$ and $CE=9$ are accepted, and $AB=10$, $BD=11$, $SC=14$, $AD=15$ are rejected because each would close a cycle, which leaves 5 edges for 6 vertices of total weight 33.
The shortest-route tree uses SA, SB, BC, BD, CE with total 38, so the cheapest network and the shortest routes differ.

4 / WHICH QUESTIONS ARE EASY

Fast methods exist for some problems and none is known for others

question	problem	status
route through every edge once	Euler circuit	degree test, fast
route through every vertex once	Hamilton cycle	NP-complete (Karp, 1972)
shortest route between two vertices	Dijkstra	fast
shortest tour through every vertex	travelling salesman	no fast method known
cheapest way to link every vertex	minimum spanning tree	Kruskal or Prim, fast
draw without crossings	planarity	linear time (Hopcroft and Tarjan, 1974)
two colours	bipartite test	linear time
three colours	3-colouring	NP-complete

5 / TRY THESE

Compute by hand

(a) What is the shortest route from S to D? S, B, D with length 20, since S, A, D costs 22.
(b) Why can the cheapest network never contain a cycle? Removing the dearest edge of a cycle leaves everything connected and cheaper.
(c) How many edges has a spanning tree on 6 vertices? 5.

ASIDE / THE TRAVELLING SALESMAN

The salesman wants a Hamilton cycle of least total weight, which is a tour and not a tree and has no fast method known, while the spanning tree gives a useful lower bound.

LINKS TO OTHER SHEETS

The Euler and Hamilton sheets give the two route problems; the Laplacian sheet counts the trees that Kruskal chooses among.

Convention. S the start vertex; weights are costs on edges; NP-complete problems have no known fast general method.